

consider $a_k \cdot p$ where $p \in \mathbb{R}$

and $\lceil a_k \cdot p \rceil$ is linearly scaled until
 $k=i$ where $\lceil a_i \cdot p \rceil \Rightarrow \lfloor a_{k+2} \cdot p \rfloor$

Consider $y=f(x): f(a)=k$ where $k \in \mathbb{R}$

$$\text{Let } f(x) \approx \hat{f}(\hat{x}): \hat{y} = \frac{(x-x_1)y_2 + (x_2-x)y_1}{x_2-x_1}$$

Generalise to $\hat{y} = a_1 L(x) + a_2 L(x)$

Assume general equation $\hat{y} = \lim_{p \rightarrow \infty} \sum_{i=1}^p \left[a_i \cdot \prod_{j=1, j \neq k}^p \left(\frac{x-x_j}{x_k-x_j} \right) \right]$